

RUBBLEJACK · REQUIREMENTS

RubbleJack requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rubblejack		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status of every requirement from RBJ-CAL-001 and the constructable design
0.2	2026-10-03	Amish Chadha	TRL 2; targets made measurable, R11 (strength margin) and R12 (cribbing) added from RBJ-DDR-001, cost requirement stated against the value-engineering target
0.1	2026-09-30	Amish Chadha	Initial scaffold

RubbleJack requirements

Every requirement is checked on paper at TRL 3 in RBJ-CAL-001; the status column gives the result. Measured verification is TRL 4 work. All requirements apply at the pump's 700 bar relief pressure, the highest pressure any part can see.

Safety: These are design targets for a safety-critical kit. Meeting them on paper does not make the kit safe to use. Every tool must be proof-loaded and inspected before any use, and the kit is not certified to NFPA 1936, NFPA 1960 or EN 13204.

Table 1. Requirements and their TRL 3 status.

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 result (RBJ-CAL-001)	Status
R1	Lifting capacity	Lift at least 10 t (22,000 lb) with the ram	CalRig proof load at 1.5 times rated load	14.5 t at 700 bar; cylinder rated 15 t	Met on paper
R2	Spreading force	At least 40 kN (9,000 lbf) between the jaw tips over the whole opening	Load cell between the tips at five openings	48.6 kN closed to 64.1 kN open	Met on paper
R3	Spreading opening	At least 250 mm (10 in) of tip opening	Measurement	262 mm open; tips 32 mm thick closed	Met on paper
R4	Cutting	Cut 16 mm (5/8 in) steel reinforcing bar, B500B or ASTM A615 grade 60	Cut test on sample bar	About 105 kN needed, 142 kN available, ratio 1.36	Met on paper
R5	Pressure safety	Every pressurised part rated at or above the 700 bar relief; hoses at least 4 to 1 burst to working	Parts rating review and relief valve test	All parts specified at 700 bar; hoses 2,800 bar burst	Met by specification; ratings to confirm when parts are bought
R6	Portable	No single packed load over 25 kg (55 lb)	Weigh each packed load	15.9, 24.8, 22.0 and 23.7 kg	Met on paper; the spreader has 0.2 kg in hand
R7	Fast setup	Pump connected and a tool working within 3 minutes of arrival	Timed drill	Two coupler connections, no tools; estimated 1 to 2 min	Met on paper
R8	Tool change	Swap between spreader, cutter and ram in under 1 minute	Timed changeover	One coupler off and one on, about 20 s	Met by design
R9	Commodity parts	Pump, hoses, couplers, gauge and cylinders all standard catalogue parts from at least two suppliers	BOM review	All in a class at least two makers offer	Met by specification
R10	Value engineering	Estimated cost reported against the USD 3,500 value-engineering target	Costed BOM	USD 5,895	Over the value-engineering target by USD 2,395
R11	Strength margin	Every made load-bearing part at least 1.5 on minimum yield at 700 bar	Strain-gauged proof load on CalRig	Lowest 1.60, the jaw arm tail at the link slot	Met on paper
R12	Lifting on cribbing	The ram never lifts on bare rubble: base plate on hardwood cribbing; ground bearing under the sleepers 1 MPa or less	Drill with the co-design partner	0.79 MPa on three 100 mm sleepers 600 mm long	Met by design and procedure

■ Requirements not met

None fails on paper. R10 is reported against the value-engineering target, not as a failure: the estimate is USD 2,395 over it, mainly because each tool has its own cylinder and the made parts use high-strength steel (RBJ-DEC-001, Value engineering). R6 is met with little margin on the spreader (0.2 kg).

■ Assumptions

- Commodity 700 bar hand pumps and 15 t cylinders are available in target regions from at least two makers (checked against catalogues, not against regional stock).
- Local workshops can waterjet or laser cut, machine and have parts heat-treated to the stated hardness.
- Crews use hardwood cribbing, which is widely available, and receive basic training with the kit.
- Strength figures use minimum catalogue yield values and conservative hand calculations; proof loads at TRL 4 confirm them.